Dictionaries and records

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

THE CONNECTED WEEK

One connected dictionary-and-record record.

Explain keys, trace an update, repair wrong-case access and distinguish missing from zero.

Keep one habit: Keep record identity, field name/type, update, membership and schema evidence together.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Return values and local state. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Lists and indexing. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Boolean conditions. Give an example and a mistake that the idea helps you avoid.

Return values and local state

> **My remembered idea, example and correction**

Lists and indexing

> **My remembered idea, example and correction**

Boolean conditions

> **My remembered idea, example and correction**

W1 · Explain the procedure

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Separate records: Blue capacity 12/used 4→used 6 gives remaining 8 then 6. Red capacity 9/used 2→used 9 gives remaining 0, capacity stays 9. Keys are case sensitive: Used is not used. Schema requires used<=capacity; capacity 3/used 4 violates it despite arithmetic -1. Shelf count 0 is present; absent price get returns None.

  Explicit priceNone is also possible but membership True; get alone does not distinguish presence. Selected describe policy treats absent/None as unavailable, not zero.

- For {"name": "Red", "capacity": 9, "used": 2}, state the keys, value of used and remaining spaces.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Predict, trace and observe

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Separate records: Blue capacity 12/used 4→used 6 gives remaining 8 then 6. Red capacity 9/used 2→used 9 gives remaining 0, capacity stays 9. Keys are case sensitive: Used is not used. Schema requires used<=capacity; capacity 3/used 4 violates it despite arithmetic -1. Shelf count 0 is present; absent price get returns None.

  Explicit priceNone is also possible but membership True; get alone does not distinguish presence. Selected describe policy treats absent/None as unavailable, not zero.

- Change the Red **record** used to 9. State remaining spaces and whether capacity changes.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Find the first error

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Separate records: Blue capacity 12/used 4→used 6 gives remaining 8 then 6. Red capacity 9/used 2→used 9 gives remaining 0, capacity stays 9. Keys are case sensitive: Used is not used. Schema requires used<=capacity; capacity 3/used 4 violates it despite arithmetic -1. Shelf count 0 is present; absent price get returns None.

  Explicit priceNone is also possible but membership True; get alone does not distinguish presence. Selected describe policy treats absent/None as unavailable, not zero.

- The code reads **record**["Used"] from the Red **record**. Explain the error and repair it without adding a field.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · State an honest limit

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Separate records: Blue capacity 12/used 4→used 6 gives remaining 8 then 6. Red capacity 9/used 2→used 9 gives remaining 0, capacity stays 9. Keys are case sensitive: Used is not used. Schema requires used<=capacity; capacity 3/used 4 violates it despite arithmetic -1. Shelf count 0 is present; absent price get returns None.

  Explicit priceNone is also possible but membership True; get alone does not distinguish presence. Selected describe policy treats absent/None as unavailable, not zero.

- A **record** has capacity 3 and used 4. It gives -1 spaces without an error. Should it be accepted? Explain.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected independent task

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Predict the three **output**s and trace the named-field update. Explain why zero and a missing count produce different results. If the condition becomes if not count, which supplied **record** is first handled incorrectly?

- Attempt this case before feedback. **Record** any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Read, update and pass a dictionary record

```
def describe(record):
    count = record.get("count")
    if count is None:
        return "missing"
    return count
record = {"count": 0}
print(describe(record))
record["count"] = 3
print(describe(record))
print(describe({}))
```

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep code and evidence

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Add a labelled dictionaries and **record**s **evidence** sheet to Python readiness portfolio. Show how you can read and update named **dictionary** fields.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this **record** to last week's **record**; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Use W2: Trace an update for Dictionaries and **record**s. **Record** the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Use the worked case for Dictionaries and **record**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Additional practice · Earlier W4 transfer

REFERENCE · SECTION 8

Additional practice · Earlier W4 transfer

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Create a Green **record** with capacity 15 and used 6. Pass its fields to remaining and store the result. Update used to 10 and repeat. Predict both results.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- A zero field is still present. A **dictionary record** describes a shelf. The program distinguishes missing data from an actual zero.

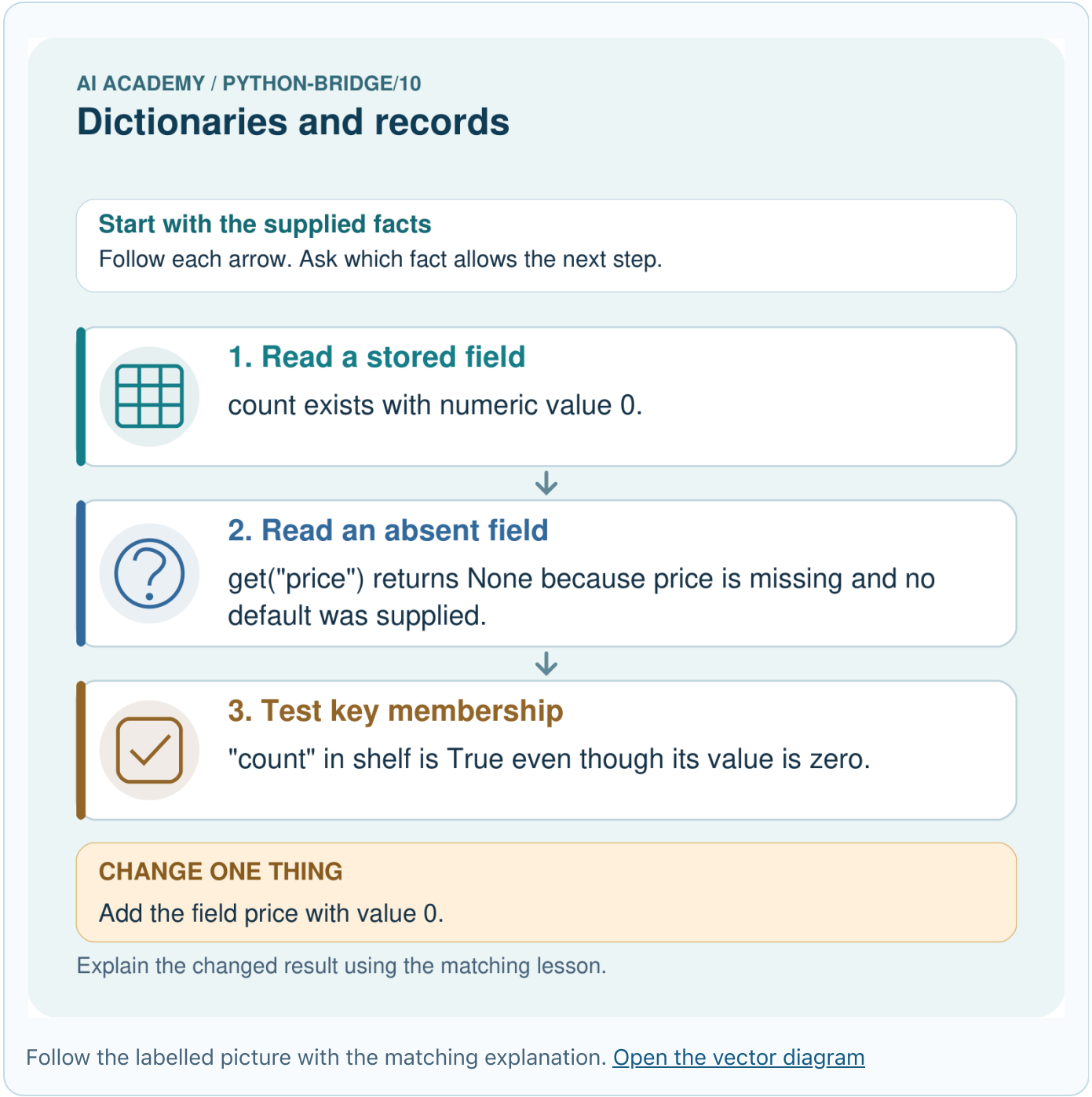
- Python 3: shelf={"count":0,"name":"Atlas"}; print(shelf["count"]); print(shelf.get("price")); print("count" in shelf).

- Predict all three original **output**s. Explain the difference between count=0 and missing price, then predict the lookup after price=0 is added.

- Support used: alone / hint / worked example. This **record** is practice **evidence**.

- Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation.

  Use actual Python 3 results to check the **prediction** after reasoning first.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project portfolio milestone**

REFERENCE · SECTION 13

Reference · Project portfolio milestone

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Project: Python readiness portfolio
- Current milestone: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a function on changed **input**s.
- For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice · Predict, check and repair**

REFERENCE · SECTION 14

Extra practice · Predict, check and repair

> Use only after an attempt; keep the reserved assessment separate.

First question and hint

- A zero field is still present. A **dictionary record** describes a shelf. The program distinguishes missing data from an actual zero.
- Which labelled step matches this detail? "count" in shelf is True even though its value is zero.
- 1. Test key membership
- 2. Read an absent field
- 3. Read a stored field

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or **record**s it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Test key membership "count" in shelf is True even though its value is zero. **Dictionary** keys name fields.

Presence is different from truthiness of a value; a measured zero should not be discarded as if the field were absent.

Changed-case question and hint

- Changed-case practice: Add the field price with value 0. Which conclusion is supported by the supplied facts?
- 1. The price lookup now returns 0 instead of None.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: The price lookup now returns 0 instead of None. A **record**ed zero is a present value, whereas the earlier lookup had no field.

Repair and keep the record

Review Return values and local state (python-bridge/9). Goal: Use a returned value in a later calculation. Short practice: Trace the two returned values and explain why first stays seven.

What happens if the caller tries to print result directly?. Return to this week and explain what you changed.

Review Lists and indexing (python-bridge/7). Goal: Distinguish a list index from the value stored at that position. Short practice: Label the three index positions.

Predict all original **output**s, then name the first **output** and stopping point after using index 3.. Return to this week and explain what you changed.

Review Boolean conditions (python-bridge/4). Goal: Translate stated requirements into comparisons. Short practice: Give both comparison results for ages 11, 15 and 16.

Which comparison rejects each outside value, and why does and matter?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Use the picture relevant to a current gap.

01. Dictionaries and records: the case

AI ACADEMY / PYTHON-BRIDGE/10 · 01 · LOOK AT THE FACTS

Dictionaries and records: the case

The situation

A zero field is still present. A dictionary record describes a shelf. The program distinguishes missing data from an actual zero.

Supplied evidence

Python 3: shelf={"count":0,"name":"Atlas"}; print(shelf["count"]); print(shelf.get("price")); print("count" in shelf).

Trace dictionaries and records and explain the changed program before running it.

Follow the labelled picture with the matching explanation. Open the vector diagram

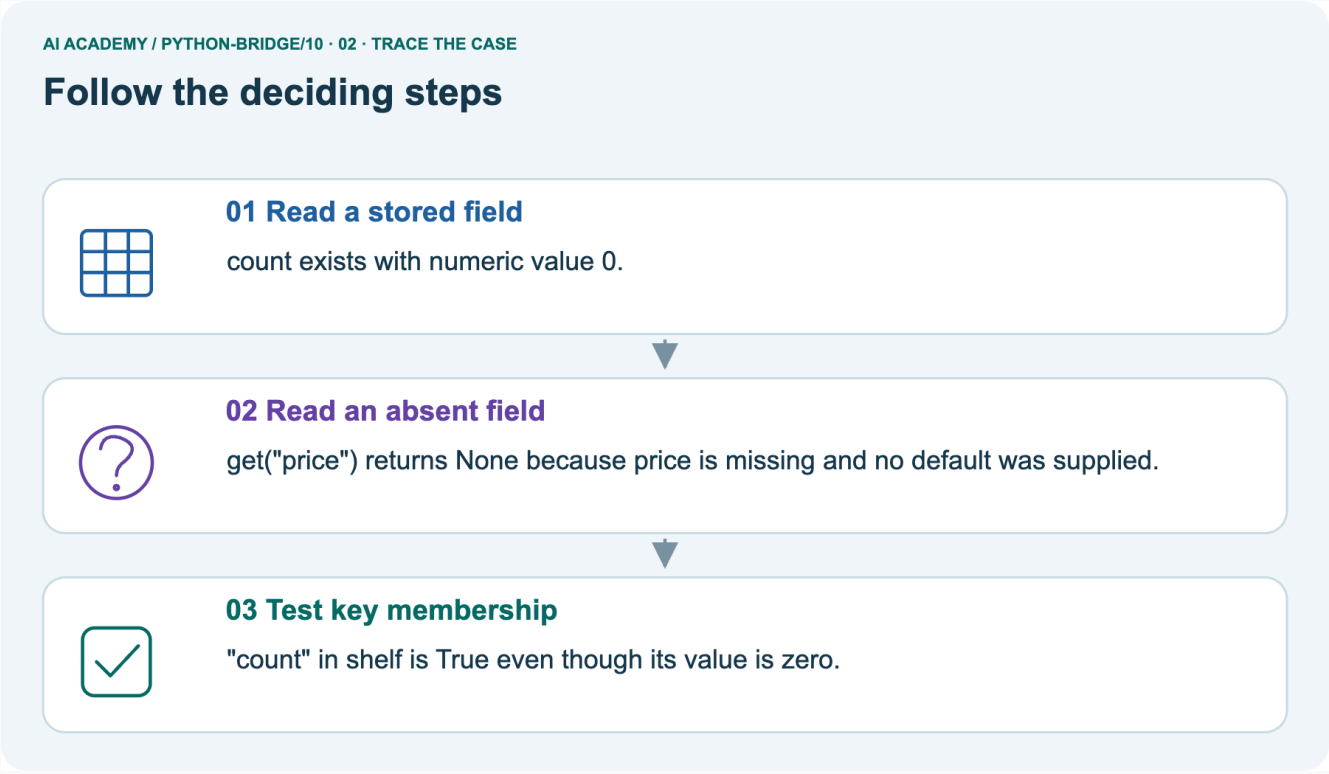
A zero field is still present. A **dictionary record** describes a shelf. The program distinguishes missing data from an actual zero.

Supplied facts: Python 3: shelf={"count":0,"name":"Atlas"}; print(shelf["count"]); print(shelf.get("price")); print("count" in shelf).

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read a stored field: count exists with numeric value 0. → Read an absent field: get("price") returns None because price is missing and no default was supplied. → Test key membership: "count" in shelf is True even though its value is zero.

Point to each numbered stage and explain which **record**ed **input** it uses.

My picture explanation, deciding evidence and one limit

03. Follow the evidence

AI ACADEMY / PYTHON-BRIDGE/10 · 03 · INSPECT THE LESSON MODEL

Follow the evidence

Step or case	Evidence	Meaning
Read used	4	12 - 4 = 8
Update used	6	12 - 6 = 6

Read each row from left to right and connect the evidence to the stated rule.

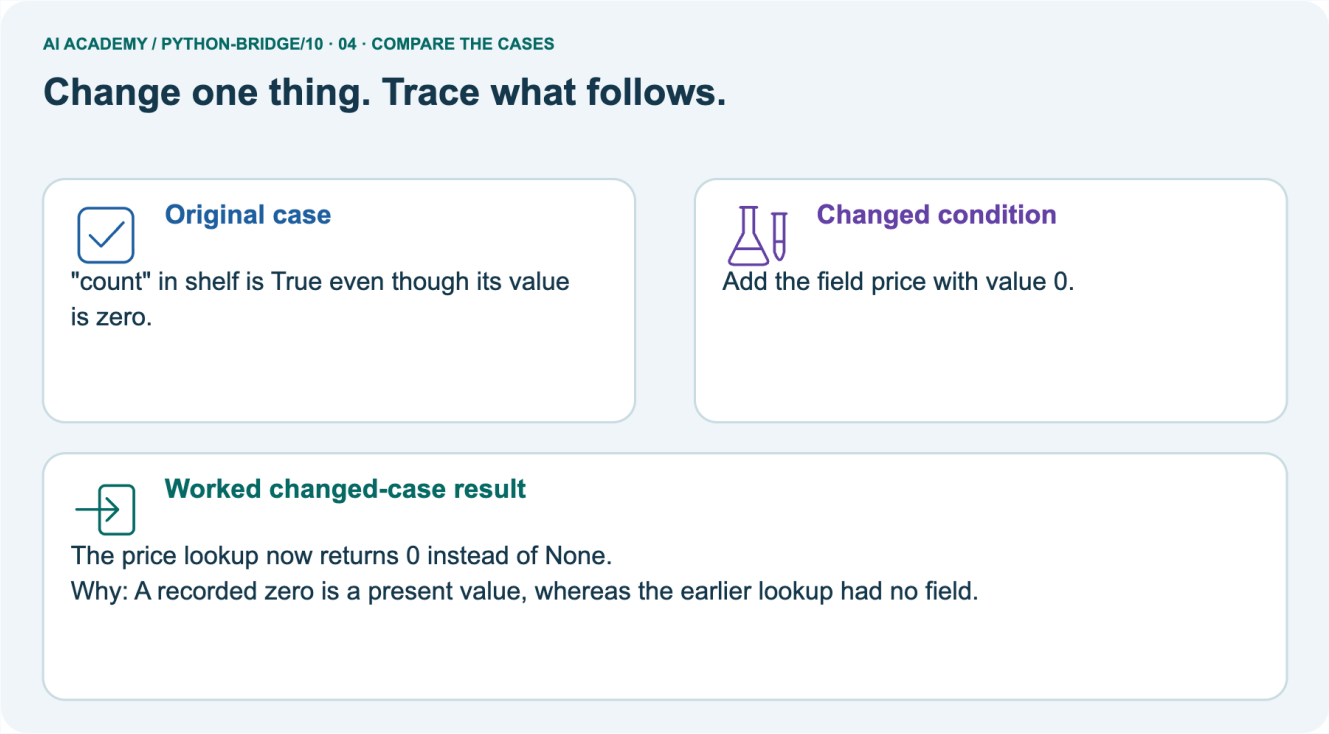
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. Read used | 4 | 12 - 4 = 8. Update used | 6 | 12 - 6 = 6

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

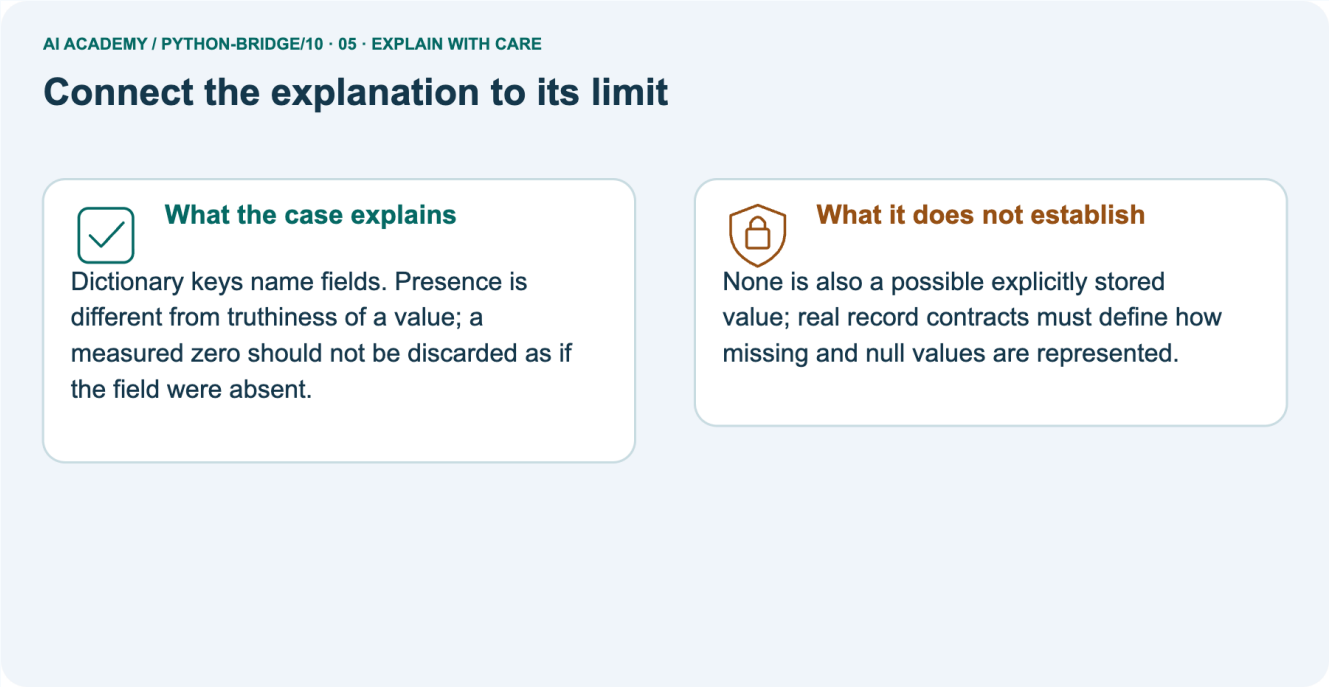
Original case: "count" in shelf is True even though its value is zero. Changed condition: Add the field price with value 0. Expected result: The price lookup now returns 0 instead of None.

Reason: A **record**ed zero is a present value, whereas the earlier lookup had no field.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

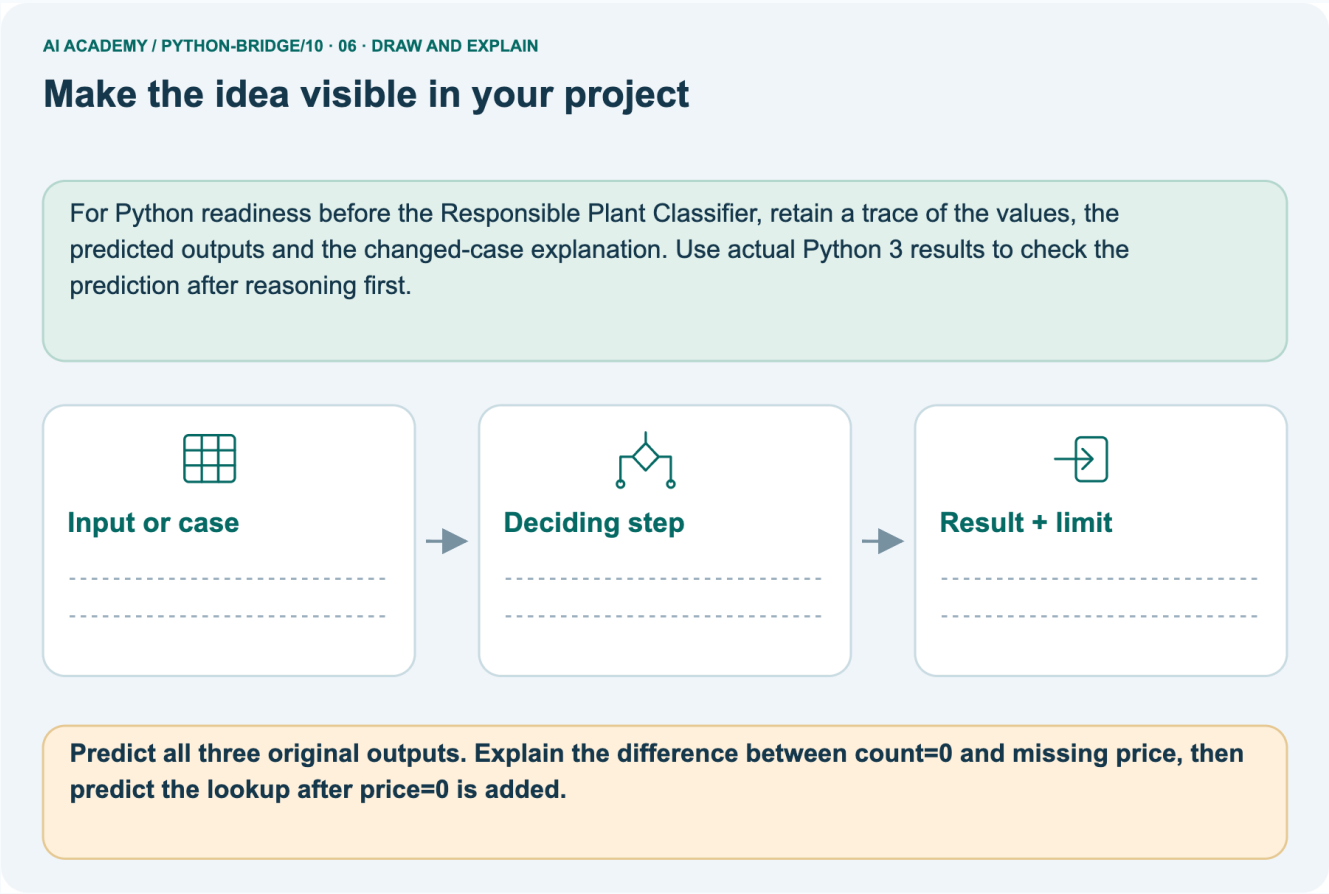
Dictionary keys name fields. Presence is different from truthiness of a value; a measured zero should not be discarded as if the field were absent.

Limit: None is also a possible explicitly stored value; real **record** contracts must define how missing and null values are represented.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Predict all three original **output**s. Explain the difference between count=0 and missing price, then predict the lookup after price=0 is added.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Dictionaries and records: the case

02. Follow the deciding steps

03. Follow the evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.